The Weinberg Angle from Three Dimensions and Two Strands

The Two Axioms Behind $\sin^2\theta_W = 1/(3\sqrt{2})$

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Abstract

This paper isolates the two structural inputs — three spatial dimensions and two rope strands — that together produce the Weinberg-angle estimate $\sin^2\theta_W = 1/(3\sqrt{2})$, and asks how much of the value is forced by them. The '3' and the '$\sqrt{2}$' each have a clear structural reading; what is not established is that these two inputs UNIQUELY force the specific combination $1/(3\sqrt{2})$ rather than merely being consistent with it. We state the estimate and this openness honestly. All numbers are outputs of the rope_solver / tests computation in this package, not inserted constants.

Scope of this paper

CLAIM: the estimate $1/(3\sqrt{2})$ rests on two structural inputs (3 dimensions, 2 strands) with clear readings.

NOT CLAIMED: that these inputs uniquely force $1/(3\sqrt{2})$. Uniqueness is Open; the ~2% gap persists. Status: Modeled (Conjecture).

1. The Two Inputs

- Three spatial dimensions — supplying the factor 3.
- Two rope strands — supplying the $\sqrt{2}$.

Combined, these give $\sin^2\theta_W = 1/(3\sqrt{2}) = 0.2357$.

2. What Is Open

The honest gap is uniqueness: the two inputs are consistent with $1/(3\sqrt{2})$, but the paper does not establish that they FORCE it — that no other combination of the same inputs is admissible. Until that is shown, and until the ~1.94% deviation from measurement is understood, the value remains a structurally-motivated conjecture rather than a derivation.

The honest status of this value. The predicted $\sin^2\theta_W = 1/(3\sqrt{2}) = 0.2357$ differs from the measured 0.23122 by about 1.94%. It is close, and its form is suggestive, but it is NOT exact, and the test suite pins it explicitly as a 'soft external input' — the value that, when fed into the lepton-mass relation instead of the measured angle, breaks it. This is therefore a geometric conjecture of the right magnitude, not a precision derivation, and the ~2% gap is the known weak link of the electroweak sector.

Status

$1/(3\sqrt{2})$ from two structural inputs — Modeled (Conjecture).

Uniqueness of the combination — Open.

~1.94% deviation — the known weak link.

Programme credit: the core Rope Hypothesis is due to Bill Gaede; this work develops and formalises it. Computational collaboration and drafting by Claude (Anthropic). Correspondence to palmer100@gmail.com.